

Binary Intelligent Reflecting Surfaces Assisted OFDM Systems

Group Meeting

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IRS and Problem Description

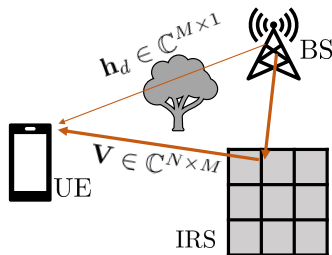


Figure: IRS setup

- Intelligent Reflecting Surfaces (IRS) are used to alter the wireless channel characteristics
- This can be leveraged to enhance the performance of existing systems

Problem: For an IRS assisted OFDM network, determine an **IRS configuration protocol** that maximizes the **sum-rate** at a given user

Solution: **Two stages:**

- 1 Channel estimation of all links - using low pilot overhead schemes
- 2 IRS phase optimization (sampled from $\{-1, +1\}$) - using MM approach

Signal Model

- IRS equipped with N elements. Phase config: $\theta \in \{-1, +1\}^N$
- OFDM system with K sub-carriers
- Time domain channel is an M -tap FIR filter

The downlink received pilot in time domain:

$$y[n] = \sum_{\ell=0}^{M-1} h_{\theta}[\ell]x[n-\ell] + w[n].$$

In frequency domain :

$$\bar{\mathbf{y}} = \bar{\mathbf{h}}_{\theta} \odot \bar{\mathbf{x}} + \bar{\mathbf{w}},$$

where $\bar{\mathbf{h}}_{\theta} \in \mathbb{C}^{K \times 1}$ is the OFDM composite channel vector with IRS phase configuration, θ and $\bar{\mathbf{w}}$ is AWGN

Channel Model

Model the ℓ th tap of the time domain channel as $h_\theta[\ell] = h_d[\ell] + \mathbf{v}_\ell^T \boldsymbol{\theta}$, where $h_d[\ell]$ is the direct path, $\mathbf{v}_\ell \in \mathbb{C}^N$ is the cascaded channel vector.

$$\bar{\mathbf{h}}_\theta = \mathbf{F} \begin{bmatrix} h_d[0] + \mathbf{v}_0^T \boldsymbol{\theta} \\ \vdots \\ h_d[M-1] + \mathbf{v}_{M-1}^T \boldsymbol{\theta} \end{bmatrix} = \underbrace{\mathbf{F} \mathbf{h}_d}_{\triangleq \bar{\mathbf{h}}_d} + \underbrace{\mathbf{F} \mathbf{V}^T \boldsymbol{\theta}}_{\triangleq \bar{\mathbf{V}}^T \boldsymbol{\theta}}$$

Direct path:

$$\mathbf{h}_d = \sum_{\ell=1}^{L_d} \beta_{d,\ell} \begin{bmatrix} \text{sinc}(0 + B(\eta - \tau_{d,\ell})) \\ \vdots \\ \text{sinc}(M-1 + B(\eta - \tau_{d,\ell})) \end{bmatrix}$$

Cascaded path:

$$\mathbf{V} = \sum_{\ell_1=1}^{L_a} \sum_{\ell_2=1}^{L_b} \beta_{r,\ell_1,\ell_2} (\mathbf{a}(\phi_{\ell_1}) \odot \mathbf{a}(\varphi_{\ell_2})) \begin{bmatrix} \text{sinc}(0 + B(\eta - \tau_{r,\ell_1,\ell_2})) \\ \vdots \\ \text{sinc}(M-1 + B(\eta - \tau_{r,\ell_1,\ell_2})) \end{bmatrix}^T$$

Phase I: Channel Estimation

Step I: Direct channel estimation

$$\bar{\mathbf{y}}_1 = \mathbf{F}(\mathbf{h}_d + \mathbf{V}^T \mathbf{1}_N) \odot \bar{\mathbf{x}} + \bar{\mathbf{w}}_1,$$

$$\bar{\mathbf{y}}_2 = \mathbf{F}(\mathbf{h}_d - \mathbf{V}^T \mathbf{1}_N) \odot \bar{\mathbf{x}} + \bar{\mathbf{w}}_2$$

The ML estimate of the direct path in frequency domain:

$$\hat{\mathbf{h}}_d = \left(\frac{\bar{\mathbf{y}}_1 + \bar{\mathbf{y}}_2}{2} \right) \oslash \bar{\mathbf{x}}$$

Step II: Cascaded channel estimation - For Q pilot transmissions,

$$\bar{\mathbf{Y}} = (\bar{\mathbf{H}}_d + \mathbf{FV}^T \boldsymbol{\Theta}) \odot \bar{\mathbf{X}} + \bar{\mathbf{W}},$$

where $\bar{\mathbf{H}}_d = \bar{\mathbf{h}}_d \otimes \mathbf{1}_Q^T$. After matched filt. and removing direct path contribution,

$$\bar{\mathbf{Z}} = \mathbf{FV}^T \boldsymbol{\Theta} + \bar{\mathbf{N}} = \bar{\mathbf{V}}^T \boldsymbol{\Theta} + \bar{\mathbf{N}},$$

where $\bar{\mathbf{Z}} \triangleq (\bar{\mathbf{Y}} \oslash \bar{\mathbf{X}}) - (\hat{\mathbf{h}}_d \otimes \mathbf{1}_Q^T)$

Cascaded channel estimation contd...

Exploit the row-sparsity in the angular domain of $\bar{\mathbf{V}}$ across the subcarriers,

$$\mathbf{Y}_{\text{sens}} \triangleq \bar{\mathbf{Z}}^T = \underbrace{\boldsymbol{\Theta}^T \mathbf{D}_{\tilde{\mathbf{A}}}}_{\triangleq \mathbf{A}_{\text{sens}}} \tilde{\mathbf{r}} + \underbrace{\bar{\mathbf{N}}^T}_{\triangleq \mathbf{N}_{\text{sens}}}$$

To estimate $\bar{\mathbf{V}}$, consider the compressed sensing problem:

$$\begin{aligned} \min_{\tilde{\mathbf{r}}} \quad & \sum_{j=1}^d \mathbf{1}\{\|\tilde{\mathbf{r}}_{j,:}\|_0 = K\}, \\ \text{s.t.} \quad & \mathbf{Y}_{\text{sens}} = \mathbf{A}_{\text{sens}} \tilde{\mathbf{r}} + \mathbf{N}_{\text{sens}} \end{aligned} \tag{P0}$$

Solve (P0) using [simultaneous orthogonal matching pursuit](#)¹

Let $\mathcal{S}^{\text{fin}}$ and $\tilde{\mathbf{r}}^{\text{fin}}$ be the estimated row support and the path gains, respectively. Then the cascaded channel estimate is

$$\hat{\mathbf{V}} = [\mathbf{D}_{\tilde{\mathbf{A}}}]_{:,\mathcal{S}^{\text{fin}}} \tilde{\mathbf{r}}^{\text{fin}}$$

¹A. Majumdar, Compressed sensing for engineers. CRC Press, 2018.

Phase II: Phase Optimization

The sum-rate is given by

$$R = \frac{B}{K+M-1} \sum_{k=0}^{K-1} \log_2 \left(1 + \frac{P|\bar{\mathbf{h}}_{\theta}[k]|^2}{BN_o} \right)$$

Let $\tilde{\mathbf{h}}_k = [\bar{h}_d[k], \bar{\mathbf{v}}_k^T]^T \in \mathbb{C}^{(N+1) \times 1}$, $\tilde{\boldsymbol{\theta}} = [1, \boldsymbol{\theta}^T]^T$, $\rho \triangleq \frac{P}{BN_o}$, and $\mathbf{h}_k \triangleq \sqrt{\rho} \tilde{\mathbf{h}}_k$. We then formulate the following problem:

$$\begin{aligned} \max_{\tilde{\boldsymbol{\theta}}} \quad & R = \frac{B}{K+M-1} \underbrace{\sum_{k=0}^{K-1} \log_2 (1 + \tilde{\boldsymbol{\theta}}^H \mathbf{h}_k \mathbf{h}_k^H \tilde{\boldsymbol{\theta}})}_{\triangleq R(K)}, \\ \text{s.t.} \quad & \tilde{\boldsymbol{\theta}} \in \{-1, +1\}^{N+1}, \quad \tilde{\boldsymbol{\theta}}[1] = 1 \end{aligned} \tag{P1}$$

- Constrained optimization problem to maximize the sum-rate over all phase configurations, $\tilde{\boldsymbol{\theta}} \in \{-1, +1\}^{N+1}$
- We use the method of [majorization-minimization \(MM\)](#)

Phase II: Phase Optimization Contd...

Lemma

²Let $\mathbf{P}$ and $\mathbf{Q}$ be two $N \times N$ Hermitian matrices such that $\mathbf{Q} \geq \mathbf{P}$. Then, for any point $\mathbf{x}_0 \in \mathbb{C}^N$, $\mathbf{x}^H \mathbf{P} \mathbf{x}$ is **majorized** by $\mathbf{x}^H \mathbf{Q} \mathbf{x} + 2\Re(\mathbf{x}^H (\mathbf{P} - \mathbf{Q}) \mathbf{x}_0) + \mathbf{x}_0^H (\mathbf{Q} - \mathbf{P}) \mathbf{x}_0$ at $\mathbf{x}_0$. Here, $\Re(\cdot)$ denotes the real part of $(\cdot)$ ■

This lemma implies that, $\forall \mathbf{x} \in \mathbb{C}^{S \times 1}$, $\mathbf{x}^H \mathbf{P} \mathbf{x} \leq \mathbf{x}^H \mathbf{Q} \mathbf{x} + 2\Re(\mathbf{x}^H (\mathbf{P} - \mathbf{Q}) \mathbf{x}^{(i)}) + \mathbf{x}^{(i)H} (\mathbf{Q} - \mathbf{P}) \mathbf{x}^{(i)}$, with equality at $\mathbf{x} = \mathbf{x}^{(i)}$

Let $\mathbf{Q} = \mathbf{h}_k \mathbf{h}_k^H$, $\mathbf{P} = \mathbf{0}_{N \times N}$ and $\mathbf{x} = \tilde{\boldsymbol{\theta}}$. Note that $\mathbf{Q} \geq \mathbf{P}$. Then,

$$R^{(K)} \geq \sum_{k=0}^{K-1} \log_2(1 + 2\Re(\tilde{\boldsymbol{\theta}}^H \mathbf{h}_k \mathbf{h}_k^H \tilde{\boldsymbol{\theta}}^{(i)}) - \tilde{\boldsymbol{\theta}}^{(i)H} \mathbf{h}_k \mathbf{h}_k^H \tilde{\boldsymbol{\theta}}^{(i)})$$

Let $C_k^{(i)} \triangleq 1 - \tilde{\boldsymbol{\theta}}^{(i)H} \mathbf{h}_k \mathbf{h}_k^H \tilde{\boldsymbol{\theta}}^{(i)}$ and $\check{\mathbf{h}}_k^{(i)} \triangleq \mathbf{h}_k \mathbf{h}_k^H \tilde{\boldsymbol{\theta}}^{(i)}$. Then

$$R^{(K)} \geq \sum_{k=0}^{K-1} \log_2(C_k^{(i)} + 2\Re(\tilde{\boldsymbol{\theta}}^H \check{\mathbf{h}}_k^{(i)}))$$

²R. Ramu, Naidu, C. R. Murthy, "Construction of unimodular tight frames for compressed sensing using majorization-minimization", **Signal Processing**, 2020.

Phase II: Phase Optimization Contd...

Lemma

³For $\mathbf{Z}, \mathbf{Y} \geq 0$, $f(\mathbf{Z}, \mathbf{Y}) \triangleq \log |(\mathbf{Z}^{-1} \mathbf{Y})|$ can be lower bounded by $f(\mathbf{Z}, \mathbf{Y}) \geq -(\log |\mathbf{Z}^{(i)}| + \text{tr}((\mathbf{Z}^{(i)})^{-1}(\mathbf{Z} - \mathbf{Z}^{(i)})) + \log |(\mathbf{Y}^{(i)})^{-1}| + \text{tr}(\mathbf{Y}^{(i)}(\mathbf{Y}^{-1} - (\mathbf{Y}^{(i)})^{-1})))$ with equality at $\mathbf{Z} = \mathbf{Z}^{(i)}$ and $\mathbf{Y} = \mathbf{Y}^{(i)}$ ■

Let $\mathbf{Y} \triangleq \text{diag}(\{C_k^{(i)} + 2\Re(\tilde{\boldsymbol{\theta}}^H \mathbf{h}_k^{(i)})\}_{k \in \{0, \dots, K-1\}})$ and $\mathbf{Z} \triangleq \mathbf{I}_{K \times K}$. Then,

$$R^{(K)} \geq - \sum_{k=0}^{K-1} \frac{C_k^{(i)} + 2\Re(\tilde{\boldsymbol{\theta}}^{(i)H} \mathbf{h}_k^{(i)})}{C_k^{(i)} + 2\Re(\tilde{\boldsymbol{\theta}}^H \mathbf{h}_k^{(i)})} + \text{terms independent of } \tilde{\boldsymbol{\theta}}$$

Let $\mathbf{g}_k^{(i)} \triangleq 2\Re(\mathbf{h}_k^{(i)})$. Then, (P1) simplifies to

$$\begin{aligned} \min_{\tilde{\boldsymbol{\theta}}} \quad & \sum_{k=0}^{K-1} \left(f_k(\tilde{\boldsymbol{\theta}}) \triangleq (C_k^{(i)} + \tilde{\boldsymbol{\theta}}^{(i)T} \mathbf{g}_k^{(i)}) / (C_k^{(i)} + \tilde{\boldsymbol{\theta}}^T \mathbf{g}_k^{(i)}) \right), \\ \text{s.t.} \quad & \tilde{\boldsymbol{\theta}} \in \{-1, +1\}^{N+1}, \tilde{\boldsymbol{\theta}}[1] = 1 \end{aligned} \tag{P2}$$

³S. S. Thoota, P. Babu and C. R. Murthy, "Codebook-Based Precoding and Power Allocation for MU-MIMO Systems for Sum Rate Maximization", in **IEEE Trans. on Comm.**, Dec. 2019.

Phase II: Phase Optimization Contd...

Using the Taylor's expansion of the cost function, we further upper bound it using a rank-1 approximation. Then (P2) becomes

$$\begin{aligned} \max_{\tilde{\boldsymbol{\theta}}} \quad & \tilde{\boldsymbol{\theta}}^T \left(\underbrace{\sum_{k=0}^{K-1} \frac{\mathbf{g}_k^{(i)}}{C_k^{(i)} + \tilde{\boldsymbol{\theta}}^{(i)T} \mathbf{g}_k^{(i)}} + 1 \left\{ C_k^{(i)} \geq \|\mathbf{g}_k^{(i)}\|_1 \right\} \frac{2\tilde{\boldsymbol{\theta}}^{(i)} \|\mathbf{g}_k^{(i)}\|_2^2 (C_k^{(i)} + \|\mathbf{g}_k^{(i)}\|_1)}{(C_k^{(i)} - \|\mathbf{g}_k^{(i)}\|_1)^3}}_{\triangleq \mathbf{q}^{(i)}} \right), \\ \text{s.t.} \quad & \tilde{\boldsymbol{\theta}} \in \{-1, +1\}^{N+1}, \tilde{\boldsymbol{\theta}}[1] = 1 \end{aligned} \tag{P3}$$

Thus, in a given MM based iteration, the solution to (P3) is

$$\tilde{\boldsymbol{\theta}}^{(i+1)} = \text{sgn}((\mathbf{q}^{(i)}[1])^* \mathbf{q}^{(i)}), \tag{1}$$

where $\text{sgn}(x) \triangleq 1$ if $\text{Re}(x) > 0$, and $\text{sgn}(x) \triangleq -1$ otherwise

Overall Algorithm

Thus, (P1) : logarithmic in $\tilde{\theta} \rightarrow$ (P2) : polynomial in $\tilde{\theta} \rightarrow$ (P3) : linear in $\tilde{\theta}$

The complete procedure: For every UE,

- 1 Compute the freq. domain CSI estimate of the direct and cascaded channels between the BS and the UE
- 2 Randomly initialize a $\tilde{\theta}$ and iteratively compute (1) until the stopping criterion

KEY MERITS OF OUR WORK:

In terms of channel estimation,

- Conventional least squares (LS) requires $\mathcal{O}(N)$ training sequences, but we require $\mathcal{O}(L_a L_b)$ pilots for channel estimation - low pilot overhead

In terms of phase optimization,

- Guaranteed convergence by directly optimizing the sum rate
- Computational complexity per UE is very small, as compared to the brute force approach which requires 2^N iterations

Numerical Results

System parameters:

- No. of time domain taps, $M = 10$,
- No. of paths, $L_a = L_b = 1$, $L_d = 10$,
- No. of OFDM sub-carriers, $K = 16$,
- Total Bandwidth, $B = 10$ MHz,
- Total no. of IRS elements, $N = 256$,
- Total no. of UEs, 20

Result - 1: On channel estimation

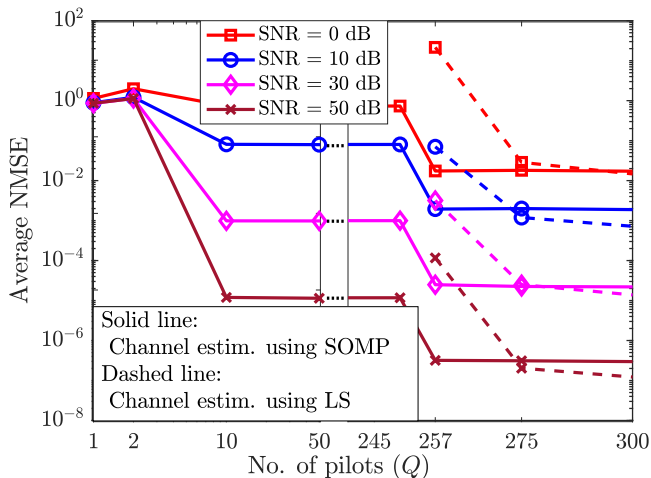


Figure: Average NMSE as a function of no. of pilots.

■ SOMP reduces pilot overhead by 96% compared to LS

Result - 2: On IRS phase optimization

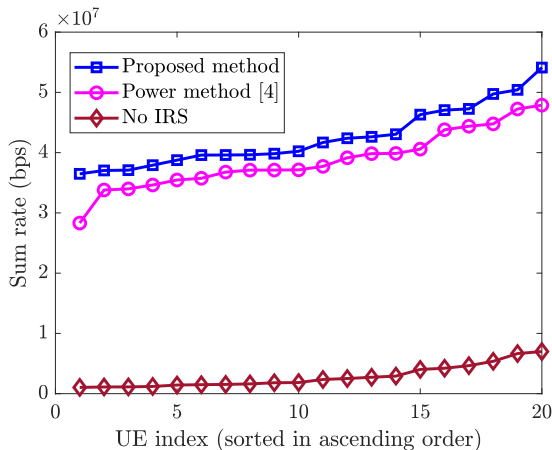


Figure: Sum rate as a function of UE index.

- Proposed method reduces the average SNR by 21% compared to literature to achieve a given rate

Conclusions

New Algorithm for phase optimization of IRS in OFDM Systems

- Exploit the joint row-sparsity and estimate the channels using low pilot overhead
- Use series of matrix inequalities and develop a MM based elegant and computationally efficient phase optimization algorithm

Thank You

Any questions?

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